

SCHEDUTOR

Automated Outbound Sales Engine

Master Engineering Plan

Architecture: Pure TypeScript + OpenClaw Skills

Target Market: Tutoring Businesses & Solo Tutors (US + Canada)

Scale: 1,000 – 10,000 leads/month

Team: Solo Founder / 1–2 Devs

Version 1.0 — April 2026

CONFIDENTIAL

Table of Contents

Table of Contents.....	2
1. Executive Summary	3
2. System Architecture	3
2.1 Architecture Overview.....	3
2.2 Technology Decisions.....	3
2.3 Project Structure.....	4
3. Database Schema.....	4
3.1 Core Tables.....	4
4. Skill Specifications	6
4.1 lead-discovery	6
4.2 lead-enrichment.....	7
4.3 email-verify	7
4.4 lead-scoring.....	7
4.5 sequence-engine	8
4.6 pipeline	8
5. Phased Delivery Plan.....	9
5.1 Phase 1: Discovery + Scoring (Weeks 1–3).....	9
5.2 Phase 2: Sequencing + Verification (Weeks 4–6).....	9
5.3 Phase 3: Pipeline + Reply Handling (Weeks 7–9).....	10
6. Operational Concerns	10
6.1 Email Deliverability	10
6.2 Legal Compliance.....	11
6.3 Rate Limiting & Politeness.....	11
6.4 Monitoring & Alerting	11
7. Cost Analysis (Monthly at Steady State)	11
8. Risks & Mitigations.....	12
9. Success Metrics.....	12
10. Immediate Next Steps (Phase 1 Kickoff)	13

1. Executive Summary

This document defines the engineering plan for Scheduler’s automated outbound sales system. The system discovers tutoring businesses and solo tutors across the US and Canada, enriches their contact information, scores them for fit, and runs personalized multi-step email sequences — all orchestrated through OpenClaw as a conversational AI interface.

Goal: Replace manual prospecting with a fully automated pipeline that finds, qualifies, and contacts 1,000–10,000 tutoring leads per month at near-zero marginal cost.

Architecture: Pure TypeScript, built as native OpenClaw skills. SQLite for persistence. Amazon SES for email delivery. Self-hosted SMTP verification. OpenClaw browser control + Google Maps Places API for lead discovery.

Timeline: Three phases over 6–9 weeks, shipping incrementally.

2. System Architecture

2.1 Architecture Overview

The system is built as a set of native OpenClaw skills that share a common SQLite database. OpenClaw’s agent runtime orchestrates all skills conversationally, and its cron system handles scheduled automation. No separate services, no inter-process communication, no additional infrastructure.

Layer	Component	Responsibility
Orchestration	OpenClaw Agent + Cron	Conversational interface, scheduling, tool routing
Skills	6 TypeScript skills	Discovery, enrichment, verification, scoring, sequencing, pipeline
Data	SQLite (better-sqlite3)	Leads, emails, sequences, pipeline state, send history
External APIs	SES, Google Maps, SMTP	Email delivery, structured place data, email verification

2.2 Technology Decisions

Decision	Choice	Rationale
Language	TypeScript	Native OpenClaw integration, single runtime, Playwright/Cheerio for scraping are mature

Database	SQLite via better-sqlite3	Zero ops, file-copy backups, handles 10k+ leads trivially, synchronous API ideal for scripts
Email Sending	Amazon SES	\$0.10/1k emails, managed deliverability, SPF/DKIM handled, no IP warming needed
Email Verification	Self-hosted SMTP check	Zero cost, ~70% accuracy, connect-to-MX + RCPT TO validation
Lead Discovery	Browser + Google Maps API	Browser for directory scraping, Maps API for structured US/CA place data
LLM (personalization)	OpenClaw's configured model	Email personalization, reply classification, lead research — uses whatever model the agent runs

2.3 Project Structure

All code lives in the OpenClaw workspace under a single skill directory with sub-modules:

```
~/ .openclaw/workspace/skills/schedutor-outbound/
├── SKILL.md                # Skill definition for OpenClaw
├── package.json            # Dependencies (better-sqlite3, cheerio, etc.)
├── tsconfig.json
├── src/
│   ├── db/                # Schema, migrations, query helpers
│   ├── discovery/         # Google Maps API, directory scrapers
│   ├── enrichment/        # Contact page scraping, email pattern matching
│   ├── verification/      # SMTP email verification
│   ├── scoring/           # Rule-based lead scoring engine
│   ├── sequencer/         # Email sequence engine + SES integration
│   ├── pipeline/          # Deal stage management + reply classification
│   └── utils/             # Shared helpers, logging, rate limiting
├── config/
│   ├── scoring-rules.json # Configurable scoring weights
│   └── sequences/         # Email sequence templates
├── data/
│   └── schedutor.db       # SQLite database file
└── tests/
```

3. Database Schema

SQLite database with the following core tables. All timestamps are ISO 8601 UTC strings. IDs are ULIDs for sortability.

3.1 Core Tables

leads

Column	Type	Description
--------	------	-------------

id	TEXT PK	ULID
business_name	TEXT NOT NULL	Company or tutor name
business_type	TEXT	'agency' 'solo_tutor' 'franchise' 'online_platform'
website	TEXT	Primary website URL
phone	TEXT	Primary phone number
address	TEXT	Physical address
city	TEXT	City
state_province	TEXT	State or province code
country	TEXT	'US' 'CA'
source	TEXT	'google_maps' 'yelp' 'directory' 'manual'
source_id	TEXT	External ID from source (e.g., Google Place ID)
score	INTEGER	Computed lead score (0–100)
pipeline_stage	TEXT	Current pipeline stage (default: 'new')
created_at	TEXT	ISO 8601 creation timestamp
updated_at	TEXT	ISO 8601 last update timestamp

lead_emails

Column	Type	Description
id	TEXT PK	ULID
lead_id	TEXT FK	References leads.id
email	TEXT NOT NULL	Email address
contact_name	TEXT	Name associated with this email
role	TEXT	'owner' 'admin' 'info' 'unknown'
verification_status	TEXT	'unverified' 'valid' 'invalid' 'catch_all' 'unknown'
source	TEXT	'scraped' 'pattern_guess' 'manual'
is_primary	INTEGER	1 if this is the preferred send address

sequences

Column	Type	Description
id	TEXT PK	ULID
name	TEXT	Sequence name (e.g., 'Agency Intro v2')
total_steps	INTEGER	Number of steps in the sequence
status	TEXT	'active' 'paused' 'archived'

sequence_steps

Column	Type	Description
id	TEXT PK	ULID
sequence_id	TEXT FK	References sequences.id
step_number	INTEGER	Order in sequence (1, 2, 3...)
delay_hours	INTEGER	Hours to wait after previous step
subject_template	TEXT	Subject line template with {{variables}}
body_template	TEXT	Email body template with {{variables}}

send_log

Column	Type	Description
id	TEXT PK	ULID
lead_id	TEXT FK	References leads.id
email_id	TEXT FK	References lead_emails.id
sequence_id	TEXT FK	References sequences.id
step_number	INTEGER	Which step was sent
ses_message_id	TEXT	SES message ID for tracking
status	TEXT	'sent' 'delivered' 'bounced' 'complained' 'opened' 'clicked'
sent_at	TEXT	ISO 8601 send timestamp

4. Skill Specifications

4.1 lead-discovery

Purpose: Find tutoring businesses and solo tutors across the US and Canada.

Data Sources: Google Maps Places API (structured), tutoring directories via browser scraping (Wyzant, Thumbtack, Tutors.com, Care.com, local directories).

Discovery Strategy:

- **Google Maps API:** Query “tutoring” + “tutoring center” + “math tutor” etc. across US/CA metro areas. Use Place Details for website, phone, address, reviews, rating. Free tier: \$200/month credit covers ~5,000 Place Details calls.
- **Directory Scraping:** Use OpenClaw’s browser tool to navigate tutoring directories. Parse listings with Cheerio. Extract business name, location, contact info, specialties.

- **Deduplication:** Match on normalized business name + city + state/province. Secondary match on phone number and domain.

Conversational Commands:

- "Find tutoring agencies in California" → Runs Google Maps API search for CA metro areas
- "Scrape Wyzant listings for Toronto" → Browser-based directory scrape
- "How many new leads this week?" → Queries SQLite for recent discoveries

4.2 lead-enrichment

Purpose: Enrich raw leads with contact emails, company size signals, and technology indicators.

Enrichment Pipeline:

- **Website Scraping:** Fetch the lead's website. Extract emails from contact pages, about pages, footer. Look for team size indicators, technology (scheduling tools they currently use), founded year, testimonials count.
- **Email Pattern Guessing:** If a website domain is found but no direct email, try common patterns: info@, hello@, contact@, firstname@, firstname.lastname@. Validate each guess with the verification skill.
- **Social Signals:** Check for LinkedIn company page, Facebook page, Google review count and rating (from Maps data). These feed into the scoring model.

4.3 email-verify

Purpose: Verify email addresses before sending to protect sender reputation.

Verification Steps:

1. DNS MX lookup — confirm the domain has mail exchange records.
2. SMTP connection — connect to the MX server, issue EHLO and MAIL FROM.
3. RCPT TO probe — check if the server accepts the recipient address.
4. Catch-all detection — test a random address at the domain to detect catch-all configurations.
5. Disposable domain check — filter known disposable email providers.

Results are stored in lead_emails.verification_status. Only "valid" emails are eligible for sequence enrollment. "catch_all" emails can be sent to but are flagged for lower confidence.

4.4 lead-scoring

Purpose: Rank leads by likelihood to convert to paying Scheduler customers.

Scoring Model (100-point scale, configurable weights):

Signal	Points	Rationale
Business type = agency	+20	Agencies have higher LTV and more complex scheduling needs
Has a website	+10	Indicates established business, more likely to adopt SaaS
Website mentions multiple tutors	+15	Multi-tutor ops need scheduling tools most
No existing scheduling tool detected	+15	Green field — no switching cost for them
Existing tool is a competitor	+5	Open to switching but harder sell
Google rating ≥ 4.0	+5	Active, quality-conscious business
Review count ≥ 20	+10	Established with active client base
Verified email found	+10	Can actually reach them
Metro area population $> 500k$	+5	Larger markets = more scheduling complexity
Has social media presence	+5	Tech-savvy, likely SaaS adopter

4.5 sequence-engine

Purpose: Send personalized multi-step email sequences and track engagement.

Sequence Architecture:

- **Templates:** Each sequence has 3–5 steps with configurable delays (e.g., Day 1, Day 3, Day 7, Day 14). Templates use `{{mustache}}` variables populated from lead data.
- **Personalization:** Before each send, the agent's LLM generates a personalized opening line based on the lead's website, specialties, and location. This is injected into the template.
- **Sending:** Emails sent via Amazon SES SDK. Rate limited to 50/hour initially during warm-up, scaling to SES account limits. Sends only during business hours (9am–5pm in the lead's timezone).
- **Tracking:** SES provides delivery, bounce, and complaint notifications via SNS webhooks. Open/click tracking via SES configuration sets.
- **Safety:** Automatic pause on bounce rate $> 5\%$. Automatic unsubscribe handling. CAN-SPAM compliant footer on all emails.

4.6 pipeline

Purpose: Track leads through the sales funnel and classify replies.

Pipeline Stages:

Stage	Entry Trigger	Exit Trigger
new	Lead discovered	Score calculated
scored	Score assigned	Enrolled in sequence
contacted	First email sent	Reply received or sequence complete
replied	Reply detected via IMAP	Reply classified by LLM
qualified	LLM classifies reply as interested	Manual: demo scheduled
demo	Demo booked (manual)	Manual: won or lost
won / lost	Manual update	Terminal states
unsubscribed	Unsubscribe link clicked	Terminal — never contact again

Reply Classification: The LLM classifies incoming replies into categories: interested (wants to learn more/schedule demo), not_interested (explicit no), question (needs info before deciding), out_of_office (auto-reply), unsubscribe (wants removal), and other. This determines the next pipeline action.

5. Phased Delivery Plan

5.1 Phase 1: Discovery + Scoring (Weeks 1–3)

Goal: Validate that you can find high-quality tutoring leads at scale before building the sending infrastructure.

Deliverables:

1. SQLite database with schema, migrations, and query helpers.
2. Google Maps Places API integration — search by metro area, extract place details, deduplicate.
3. Directory scraper for 2–3 major tutoring directories (Wyzant, Thumbtack, Care.com).
4. Basic enrichment — fetch website, extract emails from contact/about pages.
5. Lead scoring engine with configurable rules.
6. CSV export of scored leads for manual review.
7. OpenClaw SKILL.md wired up so the agent can run discovery conversationally.

Success Criteria: 500+ unique tutoring leads discovered in a single run across 5 US metros, with email addresses found for at least 40% of them.

5.2 Phase 2: Sequencing + Verification (Weeks 4–6)

Goal: Start sending personalized outbound emails with deliverability safeguards.

Deliverables:

1. SMTP email verification module.
2. Amazon SES integration with SPF/DKIM/DMARC configuration guide.
3. Sequence engine with template system and delay scheduling.
4. LLM-powered email personalization (opening lines, subject variations).
5. Cron jobs: daily verification batch, hourly send queue processing.
6. Sending safety: rate limiting, bounce rate monitoring, automatic pause.
7. CAN-SPAM compliant footer and unsubscribe link handling.

Success Criteria: First 100-lead sequence running with < 3% bounce rate and > 20% open rate.

5.3 Phase 3: Pipeline + Reply Handling (Weeks 7–9)

Goal: Close the loop — detect replies, classify intent, and manage the pipeline conversationally.

Deliverables:

6. IMAP reply monitoring (poll inbox for replies matched to sent emails).
7. LLM reply classification (interested / not interested / question / OOO / unsubscribe).
8. Pipeline stage management with automatic transitions.
9. Daily pipeline digest (summary of new replies, stage changes, key metrics).
10. Conversational pipeline queries: “Show me all qualified leads this week”, “What’s my reply rate?”
11. Enhanced enrichment: LinkedIn company page scraping, tech stack detection.

Success Criteria: End-to-end automated loop: discover → enrich → score → send → monitor replies → classify → pipeline update. First demo booked from an automated sequence.

6. Operational Concerns

6.1 Email Deliverability

This is the single highest-risk area of the project. Poor deliverability means all upstream work is wasted. Key mitigations:

- **Domain Warm-up:** Start with 20 emails/day on a fresh SES-verified domain. Increase by 20/day each week. Full volume at week 4–5.
- **Separate Domain:** Use a sending domain like mail.schedutor.com, not the primary schedutor.com domain. Protects primary domain reputation.

- **Content Quality:** LLM-personalized emails score better on spam filters than templates. Avoid spam trigger words. Keep emails short (< 150 words).
- **List Hygiene:** Verify every email before sending. Remove bounces immediately. Honor unsubscribes within 24 hours.

6.2 Legal Compliance

Cold outbound email to businesses is legal in the US under CAN-SPAM and in Canada under CASL with important differences:

- **CAN-SPAM (US):** Requires accurate sender info, clear subject lines, physical mailing address, and working unsubscribe mechanism. No opt-in required for B2B.
- **CASL (Canada):** Stricter — requires implied or express consent. B2B exception: you can send to a business address that is conspicuously published (website, directory) if the message is relevant to their business. Schedutor selling to tutoring businesses qualifies. Include unsubscribe, sender ID, and physical address.

6.3 Rate Limiting & Politeness

All scraping respects robots.txt and includes delays:

- **Google Maps API:** Stay within free tier (\$200/month). Batch requests, cache results aggressively.
- **Directory Scraping:** 2–5 second delay between page loads. Rotate user agents. Respect rate limit headers.
- **SMTP Verification:** Max 10 concurrent connections. 1-second delay between checks. Respect server timeouts.

6.4 Monitoring & Alerting

OpenClaw's cron system runs a daily health check that reports to you conversationally:

- Bounce rate trending (alert if > 3%, pause at > 5%)
- Complaint rate (alert if > 0.05%, pause at > 0.1%)
- Discovery pipeline throughput (leads/day)
- Verification success rate
- Sequence completion rates and reply rates
- Database size and growth rate

7. Cost Analysis (Monthly at Steady State)

Item	Est. Monthly Cost	Notes
------	-------------------	-------

Amazon SES	\$0.50 – \$5.00	\$0.10/1,000 emails. 5k–50k emails/month.
Google Maps API	\$0.00	\$200 free credit covers ~5k Place Details calls/month
Server (OpenClaw host)	\$5 – \$20	Small VPS or existing machine. SQLite needs minimal resources.
LLM API (via OpenClaw)	\$5 – \$30	Personalization + reply classification. Depends on model choice.
Domain + DNS	\$1 – \$2	Sending subdomain (mail.schedutor.com)
TOTAL	\$11.50 – \$57.00	Extremely low for 1k–10k leads/month pipeline

8. Risks & Mitigations

Risk	Severity	Likelihood	Mitigation
Emails land in spam	High	Medium	Domain warm-up, SES reputation management, email verification, personalized content
Google Maps API limits hit	Medium	Medium	Aggressive caching, batch by metro area, fall back to directory scraping
Directory sites block scraping	Medium	High	Multiple directory sources, browser fingerprint rotation, respect robots.txt
CASL compliance gap	High	Low	Only email conspicuously published business addresses, include all required elements
SES account suspended	High	Low	Monitor bounce/complaint rates, automatic pause thresholds, keep well under limits
OpenClaw breaking changes	Medium	Low	Pin OpenClaw version, core logic in standalone TS modules callable without OpenClaw
Low lead quality / poor ICP match	Medium	Medium	Phase 1 validates lead quality before building send infrastructure. Iterate scoring.

9. Success Metrics

The system should be evaluated against these KPIs at 90 days post-launch:

Metric	Target	Measurement
Leads discovered / month	> 2,000	COUNT from leads table
Email find rate	> 50%	Leads with verified email / total leads
Email delivery rate	> 95%	Delivered / sent from send_log
Open rate	> 25%	SES open tracking
Reply rate	> 3%	Replies detected / emails sent
Positive reply rate	> 1%	Interested replies / emails sent
Demos booked / month	> 5	Pipeline stage transitions
Bounce rate	< 3%	SES bounce notifications
Complaint rate	< 0.05%	SES complaint notifications
Cost per qualified lead	< \$5.00	Total monthly cost / qualified leads

10. Immediate Next Steps (Phase 1 Kickoff)

With this plan agreed upon, here is the exact sequence to begin Phase 1:

1. Set up the OpenClaw workspace skill directory structure as specified in section 2.3.
2. Initialize the TypeScript project with better-sqlite3, cheerio, and ulid as initial dependencies.
3. Create the SQLite schema with all tables from section 3.
4. Build the Google Maps Places API discovery module — start with 5 test metros (New York, Los Angeles, Toronto, Chicago, Vancouver).
5. Build a basic website scraper that fetches a lead's homepage and extracts email addresses.
6. Implement the scoring engine with the rules from section 4.4.
7. Wire up the SKILL.md so OpenClaw can invoke discovery conversationally.
8. Run a full discovery cycle, review the leads, and iterate on scoring weights.

Each step can be a single focused work session. At the end of Phase 1, you will have a working lead generation machine that you can review and validate before investing in the sending infrastructure.